Lecture 2

Characteristics and Specifications of Switches

There are many types of power switching devices. Each device, however, has its advantages and disadvantages and is suitable to specific applications.

The motivation behind the development of any new device is to achieve the characteristics of a "**super device**." Therefore, the characteristics of any real device can be compared and evaluated with reference to the **ideal characteristics** of a super device

Ideal Characteristics

The characteristics of an ideal switch are as follows:

1. In the on-state when the switch is on, it must have (a) the ability to carry a high forward current I_F, tending to infinity; (b) a low on-state forward voltage drop V_{ON}, tending to zero; and (c) a low on-state resistance R_{ON}, tending to zero. Low R_{ON} causes low on-state power loss P_{ON}. These symbols are normally referred to under dc steady-state conditions.

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Ideal Characteristics

2. In the off-state when the switch is off, it must have (a) the ability to withstand a high forward or reverse voltage V_{BR}, tending to infinity; (b) a low off-state leakage current I_{OFF}, tending to zero; and (c) a high off-state resistance R_{OFF}, tending to infinity. High R_{OFF} causes low off-state power loss P_{OFF}. These symbols are normally referred to under dc steady-state conditions.
3. During the turn-on and turn-off process, it must be completely turned on and off instantaneously so that the device can be operated at high frequencies. Thus, it must have (a) a low delay time t_d, tending to zero; (b) a low rise time t_r, tending to zero; (c) a low storage time t_s, tending to zero; and (d) a low fall time t_f, tending to zero.
4. For turn-on and turn-off, it must require (a) a low gate-drive power P_G, tending to zero; (b) a low gate-drive voltage V_G, tending to zero; and (c) a low gate-drive current I_G, tending to zero.

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Ideal Characteristics

5. Both turn-on and turn-off must be controllable. Thus, it must turn on with a gate signal (e.g., positive) and must turn off with another gate signal (e.g., zero or negative).
6. For turning on and off, it should require a pulse signal only, that is, a small pulse with a very small width t_w, tending to zero.
7. It must have a high dv/dt, tending to infinity. That is, the switch must be capable of handling rapid changes of the voltage across it.
8. It must have a high di/dt, tending to infinity. That is, the switch must be capable of handling a rapid rise of the current through it.
9. It requires very low thermal impedance from the internal junction to the ambient R_{JA}, tending to zero so that it can transmit heat to the ambient easily.

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Ideal Characteristics

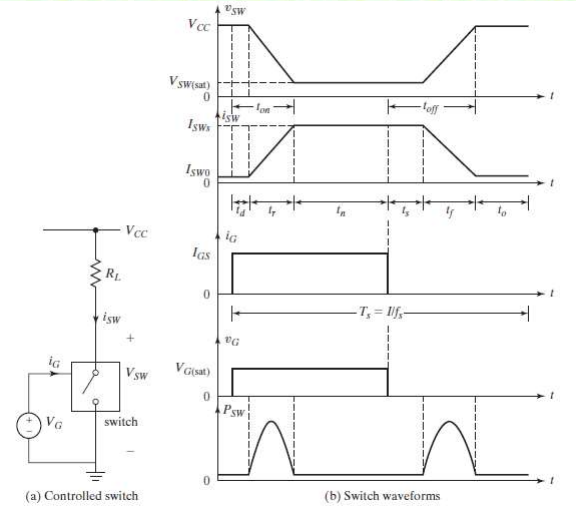
10. The ability to sustain any fault current for a long time is needed; that is, it must have a high value of i^2t, tending to infinity.
11. Negative temperature coefficient on the conducted current is required to result in an equal current sharing when the devices are operated in parallel.
12. Low price is a very important consideration for reduced cost of the power electronics equipment.

Characteristics of Practical Devices

In contrast to an ideal, lossless switch, a practical switching device dissipates some energy when conducting and switching. Voltage drop across a conducting power device is at least on the order of 1 V, but can often be higher, up to several volts. The goal of any new device is to improve the limitations imposed by the switching parameters.

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Characteristics of Practical Devices



The turn-on time (t_{on}) of a device is the sum of the delay time and the rise time, whereas the turn-off time (t_{off}) of a device is the sum of the storage time and the fall time.

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Characteristics of Practical Devices

The average conduction power loss P_{ON} is given by

$$P_{ON} = \frac{1}{T_s}\int_0^{t_n} p\,dt$$

The resultant switching power loss PSW during the turn-on and turn-off periods is given by

$$P_{SW} = f_s\left(\int_0^{t_d} p\,dt + \int_0^{t_r} p dt + \int_0^{t_s} p dt + \int_0^{t_f} p dt\right)$$

Therefore, the power dissipation of a switching device is given by:

$$P_D = P_{ON} + P_{SW} + P_G$$

The on-state power P_{ON} and the gate power P_G losses are generally low as compared to the switching loss P_{SW} during the transition time when a switch is in the process of turning on or off. The total amount of energy loss, which is the product of P_D and switching frequency f_S, could be a significant amount if the switch is operated a high frequency of kHz range.

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Switch Specifications

The characteristics of practical semiconductor devices differ from those of an ideal device. The device manufacturers supply data sheets describing the device parameters and their ratings. There are many parameters that are important to the devices.

Voltage ratings: Forward and reverse repetitive peak voltages, and an on-state forward voltage drop.

Current ratings: Average, root-mean-square (rms), repetitive peak, nonrepetitive peak, and off-state leakage currents.

Switching speed or frequency: Transition from a fully nonconducting to a fully conducting state (turn-on) and from a fully conducting to a fully nonconducting state (turn-off) are very important parameters.

$$f_s = \frac{1}{T_s} = \frac{1}{t_d + t_r + t_n + t_s + t_f + t_o}$$

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Switch Specifications

di/dt rating: The device needs a minimum amount of time before its whole conducting surface comes into play in carrying the full current. If the current rises rapidly, the current flow may be concentrated to a certain area and the device may be damaged. The di/dt of the current through the device is normally limited by connecting a small inductor in series with the device, known as a series snubber.

dv/dt rating: A semiconductor device has an internal junction capacitance C_J. If the voltage across the switch changes rapidly during turn-on, turn-off, and also while connecting the main supply, the initial current, the current $C_J dv/dt$ flowing through C_J may be too high, thereby causing damage to the device. The *dv/dt* of the voltage across the device is limited by connecting an RC circuit across the device, known as a shunt snubber, or simply snubber.

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Switch Specifications

Switching losses: During turn-on the forward current rises before the forward voltage falls, and during turn-off the forward voltage rises before the current falls. Simultaneous existence of high voltage and current in the device represents power losses. Because of their repetitiveness, they represent a significant part of the losses, and often exceed the on-state conduction losses.

Gate-drive requirements: The gate-drive voltage and current are important parameters to turn on and off a device. The gate-driver power and the energy requirement are very important parts of the losses and total equipment cost. With large and long current pulse requirements for turn-on and turn-off, the gate-drive losses can be significant in relation to the total losses and the cost of the driver circuit can be higher than the device itself.

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Switch Specifications

Safe operating area (SOA): The amount of heat generated in the device is proportional to the power loss, that is, the voltage–current product. For this product to be constant $P = vi$ and equal to the maximum allowable value, the current must be inversely proportional to the voltage. This yields the SOA limit on the allowable steady-state operating points in the voltage–current coordinates.

***I^2t* for fusing:** This parameter is needed for fuse selection. The I^2t of the device must be less than that of the fuse so that the device is protected under fault current conditions.

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Switch Specifications

Temperatures: Maximum allowable junction, case and storage temperatures, usually between 150°C and 200°C for junction and case, and between −50°C and 175°C for storage.

Thermal resistance: Junction-to-case thermal resistance, Q_{IC}; case-to-sink thermal resistance, Q_{CS}; and sink-ambient thermal resistance, Q_{SA}. Power dissipation must be rapidly removed from the internal wafer through the package and ultimately to the cooling medium. The size of semiconductor power switches is small, not exceeding 150 mm, and the thermal capacity of a bare device is too low to safely remove the heat generated by internal losses. Power devices are generally mounted on heat sinks. Thus, removing heat represents a high cost of equipment.

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